

Geometric Endpoint Statistics from Directional One-Quantum Exchange and Spatial Escape

Exact benchmark, spherical realization, and coarse-grained prediction

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Abstract

This note studies an open stochastic formation process for particles represented by nonempty lists of six directional instructions. At each motion tick one list entry is executed. While the particle remains inside a spherical source region, an incoming instruction may interact with the active entry; compatible interactions change list length by one, and formation ends when the particle crosses the boundary. No geometric endpoint law and no history weight are inserted into this spatial simulation. As an exact benchmark, a coarse-grained model with equal gain and loss rates and an independent constant escape rate is proved to assign every surviving history a weight proportional to q^L , with $0 < q < \frac{1}{2}$, and therefore to produce an exact shifted geometric endpoint law. The spatial process does not satisfy the benchmark assumptions: escape depends on position, direction, composition, and elapsed history. Nevertheless, a sweep of 30 parameter regimes and 1.8 million trajectories gives total-variation distances between empirical endpoints and best-fit geometric laws from 0.005 to 0.112. A size-resolved absorbing Markov chain, estimated on independent training trajectories, predicts held-out endpoint distributions with total-variation error from 0.0027 to 0.0152 in six tested regimes. Escape history survival is approximately exponential over a central range, but a single mean escape parameter underpredicts the observed geometric ratio. The result is therefore an approximate dynamical realization of geometric endpoint statistics, not an exact derivation for the full directional model. Temperature, the Planck spectrum, cosmological loss, and a universal speed are outside the claim.

1. Question and relation to preceding work

A preceding combinatorial note [1] proves that if every nonnegative one-quantum gain/loss history h of length $L(h)$ has weight $q^{L(h)}$, where $0 < q < \frac{1}{2}$, then its endpoint $m \geq 0$ is geometric:

$$P(m) = (1-r)r^m, \quad r = \frac{1 - \sqrt{1-4q^2}}{2q}.$$

That theorem is exact, but q^L is an ensemble definition rather than a physical mechanism. The present question is:

Can motion, directional one-quantum exchange, and escape from a finite source region generate approximately geometric endpoint sizes without manually assigning q^L to complete histories?

The answer is separated into two levels. Section 2 gives an exact constant-hazard benchmark that does generate q^L . Sections 3-8 test a richer spatial process whose escape hazard is neither independent nor constant. The benchmark explains what would be sufficient; the simulation measures how close the spatial dynamics comes to that idealization.

This paper concerns an open formation process. It is distinct from the closed reversible-exchange ensemble of a companion thermodynamic note [2]. In the closed model energy is conserved and exchange continues to equilibrium. Here a single candidate is observed when it leaves an interaction region, so escape freezes a transient endpoint.

2. Exact benchmark: symmetric gain/loss with competing escape

2.1. State and rates

Let the physical list size be $n \geq 1$ and define excess occupancy

$$m = n - 1 \geq 0.$$

At each interior state, let gain and loss occur at equal continuous-time rate $\lambda > 0$, while successful escape occurs independently at rate $\mu > 0$. A loss that would take m below zero removes the candidate from the surviving ensemble. Thus every successfully observed trajectory is a nonnegative path.

Three exponential clocks compete. Conditional on the identity of the next effective event, the probability of a specified gain or specified loss is

$$q = \frac{\lambda}{2\lambda + \mu},$$

and the probability of escape is

$$p_{\text{esc}} = \frac{\mu}{2\lambda + \mu}.$$

Since $\mu > 0$,

$$0 < q < \frac{1}{2}.$$

2.2. Complete-history weight

Consider a particular allowed history h containing L gain/loss events and then escape. Independence and the memoryless property give

$$P(h \text{ followed by escape}) = p_{\text{esc}} q^L.$$

The escape prefactor is common to all successfully completed histories. Conditioning on successful escape therefore gives relative history weight

$$w(h) \propto q^{L(h)}.$$

Applying the weighted-history theorem [1] yields

$$P(m) = (1 - r)r^m, \quad r = \frac{1 - \sqrt{1 - 4q^2}}{2q},$$

or, in terms of total size,

$$P(n) = (1 - r)r^{n-1}, \quad n \geq 1.$$

The mean excess size is

$$\langle m \rangle = \frac{r}{1 - r}.$$

Writing $x = -\log r$ gives $\langle m \rangle = \frac{1}{\exp(x) - 1}$, but this is only a change of variable. No thermal interpretation of x is made in this paper.

2.3. What the benchmark proves

The benchmark proves a conditional statement:

Equal one-quantum branching probabilities plus a state-independent constant escape hazard produce the complete-history factor q^L and an exact geometric endpoint law.

It does not prove that crossing a spatial boundary has a constant hazard, that directional trajectories of equal length are equiprobable, or that blocking a loss at $n = 1$ is equivalent to killing the candidate. Those differences are tested rather than ignored below.

3. Six-direction spatial formation model

3.1. Particle state and motion

A candidate particle is a finite nonempty list with entries in

$$\mathcal{D} = \{R, L, U, D, F, B\},$$

identified with the six signed coordinate vectors $\pm e_x, \pm e_y, \pm e_z$. The candidate begins at the center of a ball of radius R with one random instruction.

At each tick:

1. Choose one list entry uniformly and execute a taxicab step of length d .
2. If the new position lies outside the ball, record the particle and stop.
3. Otherwise, attempt an encounter with probability p .
4. Choose a source point uniformly inside the ball. Quantize the displacement from source to particle to its dominant signed coordinate axis.
5. Accept the encounter only when the incoming axis is perpendicular to the active instruction's axis.
6. On acceptance, add the incoming instruction with probability $\frac{1}{2}$; otherwise remove the active instruction if $n > 1$.

The last condition blocks loss at $n = 1$ rather than killing the candidate. The model therefore has a reflecting size boundary but an absorbing spatial boundary. This is one reason the exact benchmark cannot be transferred without qualification.

3.2. Recorded observables

For every escaped particle the simulation records:

- endpoint size n ;
- number L of successful ± 1 transitions;
- tick age t ;
- internal normalized speed $v = \frac{\|\sum_i \mathbf{d}_i\|}{n}$; and
- angle between the net instruction vector and the exit radius.

The primary observable of this paper is $P(n)$. Motion and speed are diagnostics, not part of the geometric-law claim.

3.3. No manual path factor

The simulation does not choose a history length from a geometric distribution, does not multiply trajectories by q^L , and does not stop with a fixed probability per interaction. Escape occurs only when executed taxicab motion crosses the sphere. Hence any approximate exponential survival or geometric endpoint shape is produced by the coupled motion/exchange process.

4. Fitting and error measures

For a sample of endpoints $n \geq 1$, the shifted geometric maximum-likelihood parameter is determined by the sample mean:

$$\hat{r} = \frac{|n| - 1}{|n|}.$$

The fitted probability is

$$G_{\hat{r}}(n) = (1 - \hat{r})\hat{r}^{n-1}.$$

Agreement is measured by total-variation distance

$$D_{\text{TV}} = \frac{1}{2} \sum_{n \geq 1} |P_{\text{emp}(n)} - G_{\hat{r}}(n)|.$$

$D_{\text{TV}} = 0$ means identical distributions; $D_{\text{TV}} = 1$ means disjoint support. Because $\hat{r}$ is estimated from the same endpoints, a small value demonstrates shape agreement, not an independently predicted parameter. Section 7 therefore performs a stronger held-out prediction test.

5. Robustness sweep of endpoint geometry

The implementation `test_geometric_robustness.cpp` tested five radii

$$R \in \{0.5, 1, 2, 3, 5\}$$

and six Poisson encounter intensities

$$\lambda \in \{0.03, 0.1, 0.3, 1, 3, 10\}.$$

Each of the 30 regimes used 60,000 trajectories, for 1.8 million total. The Poisson formulation allows multiple encounters per tick and is a high-flux extension of the one-attempt Vue implementation. All other directional, perpendicular, gain/loss, and escape rules are the same.

Representative results are:

R	λ	$ n $	$\hat{r}$	D_{TV}
0.5	0.03	1.136	0.119	0.0049
0.5	0.30	2.094	0.522	0.0546
1.0	0.30	2.989	0.665	0.0564
2.0	1.00	9.195	0.891	0.0791
3.0	0.30	5.620	0.822	0.0633
5.0	1.00	15.828	0.937	0.0880
5.0	10.0	60.038	0.983	0.1120

Across all 30 regimes,

$$0.0049 \leq D_{\text{TV}} \leq 0.1120.$$

The fit is strongest at low occupancy and gradually worsens in the largest, highest-flux regimes. Tail-only TV contributions were approximately 0.004 to 0.061. The result is robust approximate geometry, not an exact stationary law.

6. Are escape-history lengths exponential?

For a completed trajectory let L count only successful size transitions, not incompatible or rejected encounters. Define the survival function

$$S_{L(\ell)} = P(L \geq \ell).$$

If escape supplied a constant hazard per successful transition, the adjacent survival ratio

$$q_{\text{esc}(\ell)} = \frac{S_{L(\ell+1)}}{S_{L(\ell)}}$$

would be constant. The implementation `test_escape_history_weight.cpp` used 120,000 trajectories at $R = 3$ for each intensity. Over a central range with at least 500 surviving trajectories it measured:

λ	$ L $	R_{exp}^2	$ q _{\text{esc}}$	$\text{CV}(q)$
0.1	9.30	0.9867	0.8408	0.0572
0.3	41.61	0.9737	0.9461	0.0247
0.5	81.90	0.9604	0.9684	0.0169
1.0	200.44	0.9334	0.9849	0.0097
2.0	473.97	0.9019	0.9928	0.0054

Here R_{exp}^2 is the coefficient of determination of a linear fit to $\log S_{L(\ell)}$, and $\text{CV}(q)$ is the coefficient of variation of adjacent survival ratios. The central history-length survival is close to exponential, especially at low and moderate intensity, but curvature becomes more visible at high intensity.

The correlation R^2 between L and tick age increased from 0.725 at $\lambda = 0.1$ to 0.941 at $\lambda = 2$. Thus L becomes increasingly close to a linear clock, but it is not exactly proportional to time.

7. Why one escape parameter is insufficient

If every successful transition were an independent fair gain/loss choice, the probability of one specified branch would add a factor $\frac{1}{2}$. A tempting estimate is therefore

$$q_{\text{path}} \approx \frac{|q|_{\text{esc}}}{2},$$

followed by the exact Catalan map

$$r_{\text{path}} = \frac{1 - \sqrt{1 - 4q_{\text{path}}^2}}{2q_{\text{path}}}.$$

This estimate has the required range $q_{\text{path}} < \frac{1}{2}$, but it systematically underpredicts the endpoint parameter:

λ	q_{path}	r_{path}	r_{endpoint}
0.1	0.420	0.546	0.664
0.3	0.473	0.715	0.822
0.5	0.484	0.775	0.870
1.0	0.492	0.839	0.915
2.0	0.496	0.887	0.944

The discrepancy has a physical interpretation. Spatial escape is correlated with direction and composition; the reflecting size boundary biases gain/loss histories; and paths of equal length are not equiprobable. Approximate exponential length survival is therefore evidence for a path penalty, but not a complete prediction of $P(n)$.

8. Size-resolved absorbing-chain prediction

To retain the dominant state dependence, define a tick-scale coarse-grained kernel

$$K_{nj} = P(n_{t+1} = j \text{ and no escape} \mid n_t = n)$$

and escape probability

$$a_n = P(\text{escape on next tick} \mid n_t = n).$$

These quantities were estimated from 80,000 training trajectories. Starting with unit mass at $n = 1$, probability mass was propagated by K and absorbed through a_n . No geometric fit to the held-out endpoints was used. An independent set of 80,000 trajectories supplied the test distribution.

Results from `predict_endpoints_from_escape_chain.cpp` were:

R	λ	$\hat{r}_{\text{test}}$	geometric TV	chain TV
1	0.1	0.436	0.0538	0.0027
1	0.3	0.665	0.0588	0.0032
3	0.1	0.665	0.0666	0.0051
3	0.3	0.823	0.0633	0.0071
3	1.0	0.915	0.0877	0.0152
5	1.0	0.937	0.0960	0.0133

The size-resolved chain predicts held-out endpoints substantially better than a one-parameter geometric fit. This does not prove that size alone is a fundamental Markov state: direction and position have been averaged out. It shows that their conditional effects are represented accurately enough by an empirical size-dependent kernel for the tested regimes.

9. Speed and anisotropy diagnostics

The normalized internal speed at exit is

$$v = \frac{1}{n} \left\| \sum_{i=1}^n \mathbf{d}_i \right\|.$$

Earlier exploratory runs suggested a possible large-size value near $\frac{1}{\sqrt{3}}$. A controlled comparison using the exact one-encounter-per-tick Vue rules did not establish such a universal limit. With $R = 20$, encounter probability 0.95, and perpendicular-only exchange, mean speeds were

size range	all exits	axis-like	diagonal-like
20-49	0.659	0.714	0.591
50-99	0.623	0.671	0.566
100-149	0.586	0.627	0.535
150-249	0.543	-	-

The directional difference is substantial. Six fixed taxicab directions leave microscopic anisotropy, and averaging over exits can pass near $\frac{1}{\sqrt{3}}$ without establishing a direction-independent attractor. Perpendicularity reduced speed dispersion in separate controls but did not remove orientation dependence. Consequently neither a universal speed nor emergent rotational invariance is a result of this paper.

10. Interpretation and limitations

The numerical evidence supports the following limited chain:

Directional one-quantum exchange plus boundary crossing generates approximately exponential history survival and robust approximately geometric endpoint sizes. A size-resolved absorbing kernel predicts the residual deviations accurately in held-out simulations.

The evidence does not support stronger claims:

- The full spatial law is not exactly geometric.
- A single measured q_{esc} does not determine the endpoint ratio.
- Gain/loss probability $\frac{1}{2}$ is a modeling assumption, not a derived physical law.
- The dominant-axis source quantization retains taxicab anisotropy.
- The Poisson high-flux sweep is not identical to the browser’s Bernoulli one-encounter-per-tick implementation.
- No link from interaction intensity to thermodynamic temperature is derived.
- No Planck density-of-states factor, low-energy suppression, tired-light law, cosmological reset, or matter-creation process is included.

The spherical model is therefore best regarded as a constructive open-system example and a source of quantitative targets for future microscopic exchange rules.

11. Reproducibility

The principal programs are:

- `test_geometric_robustness.cpp` - 30-regime endpoint sweep;
- `test_escape_history_weight.cpp` - history survival and one- q test;
- `predict_endpoints_from_escape_chain.cpp` - held-out absorbing-chain test;
- `test_vue_large_n_perpendicular.cpp` - speed and anisotropy diagnostic; and
- `DirectionalSourceEscape.vue` - interactive browser realization.

The public project repository is github.com/matterasmachine/matterasmachine. A final content-bearing commit or release containing this manuscript and all five listed implementations must be cited before publication. Numerical tables in this draft correspond to the source files in the local development tree on 2026-07-03. Fixed pseudorandom seeds are embedded in the C++ programs; exact bitwise output may still depend on the standard-library implementation of random distributions.

12. Conclusion

An independent constant escape clock supplies an exact physical realization of the complete-history weight q^L and therefore an exact shifted geometric endpoint law. Spatial boundary crossing is richer: its hazard depends on position, direction, composition, and the history that brought the particle there. It does not satisfy the exact benchmark.

Even so, the directional spherical process produces a stable approximate geometric shape across a broad parameter sweep. Its history survival is close to exponential over a central range, and a size-resolved absorbing Markov chain predicts independent endpoint samples with roughly 0.3%-1.5% TV error in the six reported regimes. These observations identify escape as a plausible dynamical source of length suppression while showing why a single scalar q cannot capture the complete spatial process.

The next theoretical task is not to impose a more elaborate target distribution. It is to specify microscopic transfer rules that determine the state-dependent kernel analytically and to test whether a controlled limit reduces that kernel to the exact constant-hazard benchmark.

13. References

- [1] Matter-as-machine, “Geometric Occupancy from Weighted One-Quantum Gain/Loss Histories: Relation to Bose-Einstein and Planck Forms,” unpublished note, 2026.
- [2] Matter-as-machine, “Thermodynamic Equilibrium from Reversible One-Quantum Exchange among Discrete Particles,” unpublished note and reproducibility archive, 2026.
- [3] Y. Sun and F. Ma, “Some New Binomial Sums Related to the Catalan Triangle,” *Electronic Journal of Combinatorics* **21** (2014), Paper P1.33. doi:10.37236/3701.
- [4] V. Giorno and A. G. Nobile, “Bell Polynomial Approach for Time-Inhomogeneous Linear Birth-Death Process with Immigration,” *Mathematics* **8** (2020), 1123. doi:10.3390/math8071123.
- [5] J. R. Norris, *Markov Chains*, Cambridge University Press, 1997, chapters 1-2.

Appendix A: exact benchmark in discrete event time

The same benchmark can be written without exponential clocks. At each effective event choose gain with probability q , loss with probability q , and escape with probability $1 - 2q$, where $0 < q < \frac{1}{2}$. A particular allowed history of length L followed by escape then has probability

$$(1 - 2q)q^L.$$

Conditioning on successful escape removes the common factor $1 - 2q$, leaving the weighted-history ensemble of [1]. Continuous time determines waiting times but is not required for the endpoint theorem.

Appendix B: implementation distinctions

The interactive Vue implementation attempts at most one encounter per tick with Bernoulli probability p . The robustness and absorbing-chain C++ programs use a Poisson number of encounters per tick with mean λ , allowing multiple events at high flux. Both versions execute one motion instruction before testing escape and use the same uniform source ball, dominant-axis quantization, perpendicular compatibility, reflecting size boundary, and fair gain/loss choice. Results from the two encounter schedules should not be numerically identified without a small- p, λ scaling analysis.